IOQM 2024

Exercise 1 (IOQM 2024 P10,). Determine the number of positive integral values of p for which there exists a triangle with sides a, b and c which satisfy

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = 0$$

Summary — Complete squares and use the triangle inequality.

Walkthrough —

(a) Note that

$$a^2 + (p^2+9)b^2 + 9c^2 - 6ab - 6pbc = (a-3b)^2 + (pb-3c)^2.$$

(b) This gives $a = 3b$ and

$$c = \frac{p}{3}b.$$

(c) Use the inequality[a] $a + b > c$, to obtain

$$3b + b > \frac{p}{3}b,$$

which yields $p \leq 11$.

(d) Use the inequality[b] $b + c > a$, to get

$$b + \frac{p}{3}b > 3b,$$

implying $p \geq 7$.

(e) It reduces to finding the number of positive integers p such that $7 \leq p \leq 11$.

What about the condition $c + a > b$? Anything else to be taken care of?

[a] How does it follow?
[b] How does it follow?

Solution 1. to write ■

Exercise 2 (IOQM 2024 P15,). Let X be the set consisting of twenty positive integers $n, n+2, \ldots, n+38$. The smallest value of n for which any three numbers $a, b, c \in X$, not necessarily distinct, form the sides of an acute-angled triangle is:

Summary — Use Pythagoras' theorem.

Walkthrough —

(a) Use the fact that[a] if a, b, c are the sides of an acute-angled triangle with c denoting the largest side, then $a^2 + b^2 > c^2$ holds.

(b) Show that for any such n, the inequality

$$(n + 38)^2 < 2n^2$$

holds, which yields $n \geq 92$.

(c) Prove that for $n = 92$, the inequality

$$2(n + 2i)^2 > (n + 2i + 2)^2$$

holds for $i = 0, 1, \ldots, 18$.

(d) Note that if suffices to show that

$$2m^2 > (m + 2)^2$$

for any $m \in \{92, 94, \ldots, 128\}$.

What is the role of part (c)? Does it guarantee that for $n = 92$, the set X has the stated property?

[a] How to prove it? Does the **converse** of this statement hold? In other words, if a, b, c are positive real numbers satisfying $a^2 + b^2 > c^2$, then does it follow that a, b, c are the sides of an acute-angled triangle? Even if it doesn't, would it be true under additional hypothesis?

Solution 2. to write ■

Exercise 3 (IOQM 2024 P20,). On a natural number n you are allowed two operations:

1. multiply n by 2, or
2. subtract 3 from n.

For example starting with 8 you can reach 13 as follows: $8 \longrightarrow 16 \longrightarrow 13$. You need two steps and you cannot do in less than two steps. Starting from 11, what is the least number of steps required to reach 121?

Walkthrough —

(a) Does

$$121 \longleftarrow 124 \longleftarrow 62 \longleftarrow 31 \longleftarrow 34 \longleftarrow 17 \longleftarrow 20 \longleftarrow 10 \longleftarrow 5 \longleftarrow 8 \longleftarrow 11$$

help?

(b) Consider a sequence of operations, with least number of steps, to be applied to 11 to reach 121.

(c) Note that if m denotes the number of steps where operation (1) (that is, multiplication by 2) has been applied, then $121 \leq 2^m \cdot 11$ holds, and hence m is at least 4.

(d) Consider congruence conditions modulo 3 to show that m is odd.

(e) If each of the last three steps of reaching 121 uses the second operation, and fourth last step does not use it, then the integers obtained in the last few steps are

$$121 \longleftarrow 124 \longleftarrow 127 \longleftarrow 130 \longleftarrow 65 \longleftarrow 68 \longleftarrow \cdots \longleftarrow 11.$$

Note that to reach 68 from 11, the second operation has to be used at least once. This shows that the total number of steps is at least 10.

(f)

Solution 3. to write ■

Exercise 4 (IOQM 2024 P8,). Let n be the smallest integer such that the sum of the digits of n is divisible by 5, as well as the sum of the digits of $n+1$ is divisible by 5. What are the first two digits of n in the same order?

Summary — Consider the last digit of n. Next consider its second last digit, the third last digit etc.

Walkthrough —

(a) Show that the last digit of n is 9.

(b) Note that n has at least two digits. If n is a two-digit number, then show that it is not equal to any integer other than 19 or 69.

(c) Conclude that n has at least three digits, and its second last digit is 9.

(d) If n is a three-digit number, then show that it is not equal to any integer other than 299 or 799.

(e) Argue similarly to show that n has at least four digits, and its third last digit is 9.

(f) Prove that n has at least five digits, and its fourth last digit is 9.

(g) If n is a five-digit number, then show that it is equal to 49999.

> Do we need to check that 49999 has the required property?

Solution 4. to write ■